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Brac University
Semester: Fall 2025
Course Code: CSE251
Electronics Devices and Circuits

Assessment: *Final Exam*
Duration: 1 hour 45 minutes
Date: January 13, 2026
Full Marks: 55

✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
✓ All **5 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
✓ Symbols have their usual meanings.

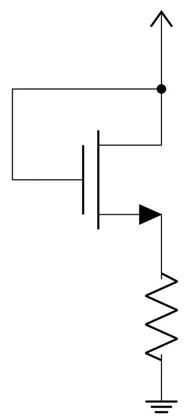
■ Question 1 of 5 *[CO1] [8 marks]*

(a) *[2 marks]* The output voltages of a full-wave and a half-wave rectifier with separate filter capacitors have the same peak and ripple voltages for the same input AC sinusoidal voltage. Explain how can you identify which voltage is from which rectifier.

(b) *[3 marks]* For each operating mode of an NPN BJT listed below, use a checkmark (✓) to indicate whether the base-emitter (BE) and base-collector (BC) junctions are Forward Biased (F.B.) or Reverse Biased (R.B.).

Operating Mode	**Base-Emitter (BE)**		**Base-Collector (BC)**	
	F.B.	**R.B.**	**F.B.**	**R.B.**
Active	☐	☐	☐	☐
Saturation	☐	☐	☐	☐
Cut-off	☐	☐	☐	☐

(c) *[3 marks]* In the following circuit, given that the gate and drain terminals are short-circuited, show that if the MOSFET is ON (i.e., a gate-to-source voltage greater than the threshold voltage), it satisfies the condition to operate in the saturation region.



■ Question 2 of 5 *[CO3] [10 marks]*

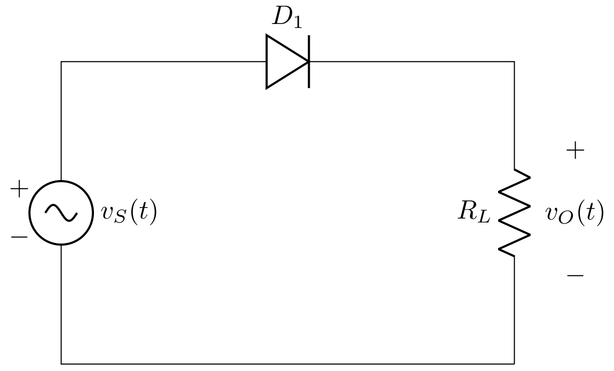
(a) *[5 marks]* A 12 V amplitude, 2 kHz sinusoidal AC voltage is rectified and filtered using a FW rectifier with 4 µF filter capacitor. The diodes used in the rectifier are identical and have a cut-in voltage of 1.2 V. If the average output voltage is measured to be 9 V, determine the value of the load resistance.

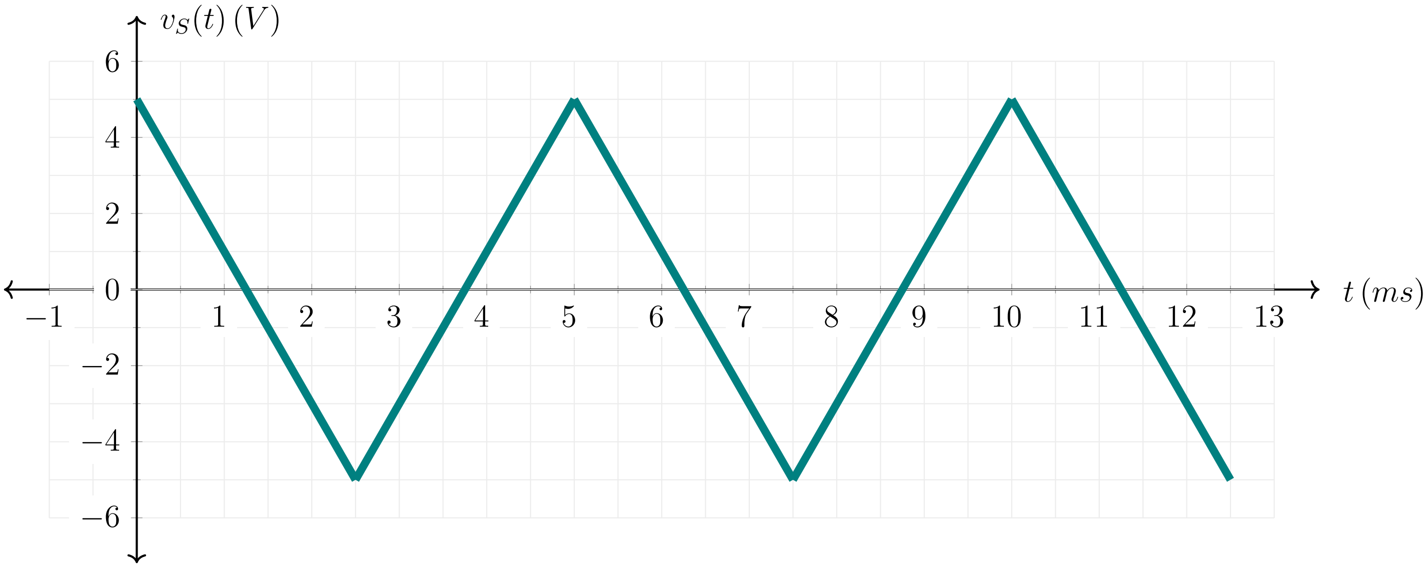
(b) *[5 marks]* Implement the following logic function using MOSFETs. Here, A, B, C, D, and E are Boolean inputs, and f is the Boolean output:

$$f = (A + B) \cdot C + D \cdot E$$

■ Question 3 of 5 *[CO2] [5 marks]*

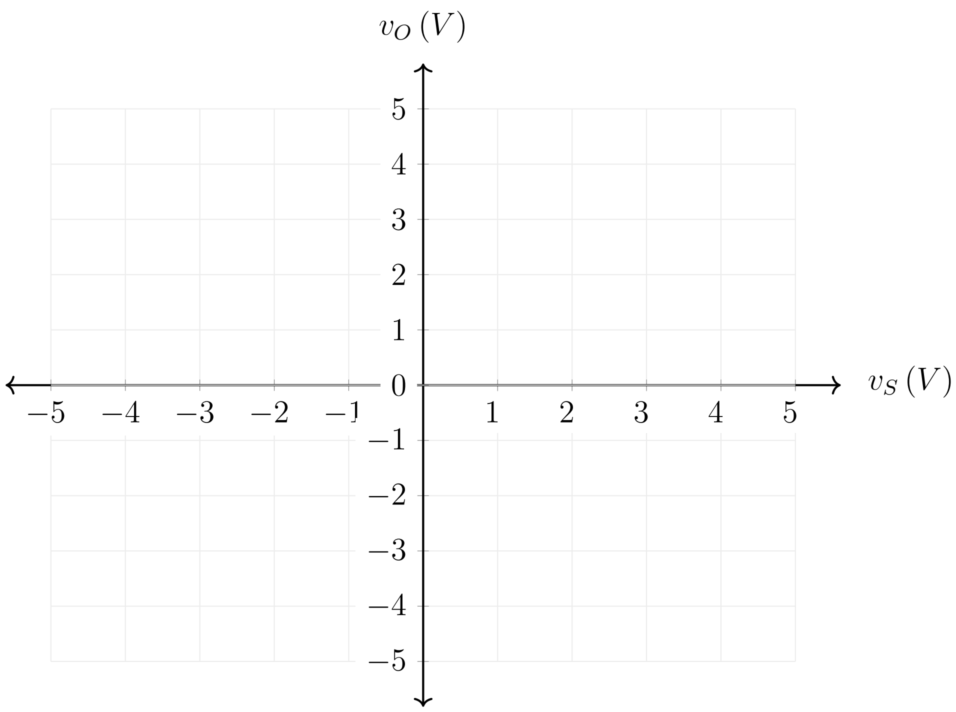
A rectifier circuit is designed to drive a resistive load R_L using a diode D_1 with cut-in voltages of 1 V, as shown in the schematic below. The circuit rectifies an AC input voltage $v_S(t)$, illustrated in the accompanying plot.





(a) ***[2 marks]*** On the same grid as $v_S(t)$, sketch approximately the waveform of the output voltage $v_O(t)$.

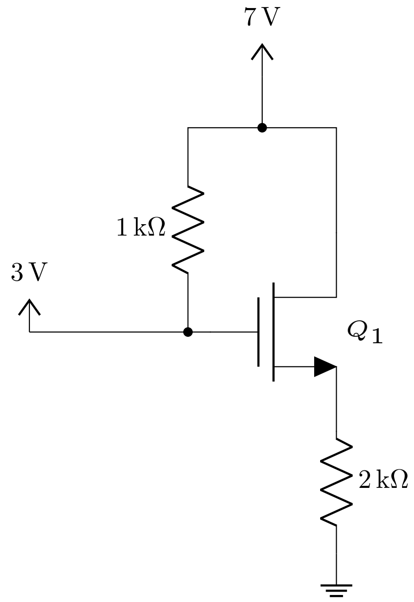
(b) ***[3 marks]*** Write an equation relating $v_O(t)$ and $v_S(t)$, and draw the voltage transfer characteristic (VTC) graph for the rectifier on the following blank grid.



■ Question 4 of 5 *[CO2] [22 marks]*

Analyze the following circuits to answer the questions below:

(a) ***[8 marks]*** The NMOS transistor Q_1 in the following circuit has parameters given in the box. Determine the current supplied by the 7 V source.



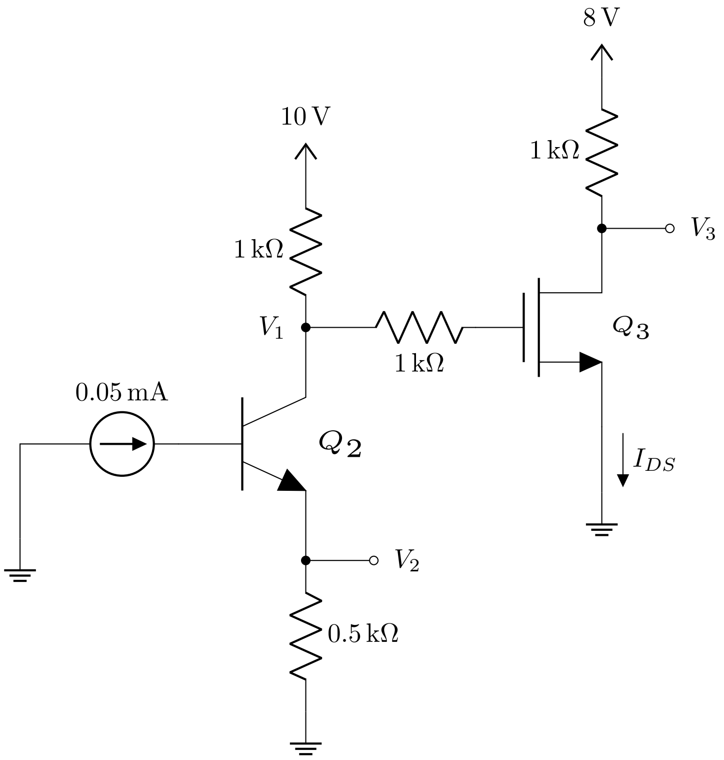
N-Channel MOSFET Parameters:

For the transistor Q_1,

Threshold voltage: $V_{T_n} = 2\,\text{V}$

Transconductance parameter: $k_n = 2\,\text{mA/V}^2$

(b) The transistors Q_2 and Q_3 in the following circuit have parameters given in the box.



NPN BJT Parameters:

For the transistor Q_2,

Common emitter current gain: $\beta = 100$

Base-emitter voltage in the active region: $V_{BE(active)} = 0.7\,\text{V}$

Base-emitter voltage in the saturation region: $V_{BE(sat)} = 0.8\,\text{V}$

Collector-emitter voltage in the saturation region: $V_{CE(sat)} = 0.2\,\text{V}$

N-Channel MOSFET Parameters:

For the transistor Q_3,

Threshold voltage: $V_{T_n} = 1\,\text{V}$

Transconductance parameter: $k_n = 2\,\text{mA/V}^2$

(i) ***[6 marks]*** Determine the values of the voltages V_1 and V_2.

(ii) ***[8 marks]*** Determine the voltage V_3 and the current I_{DS}.

N-Channel MOSFET Current Equations:

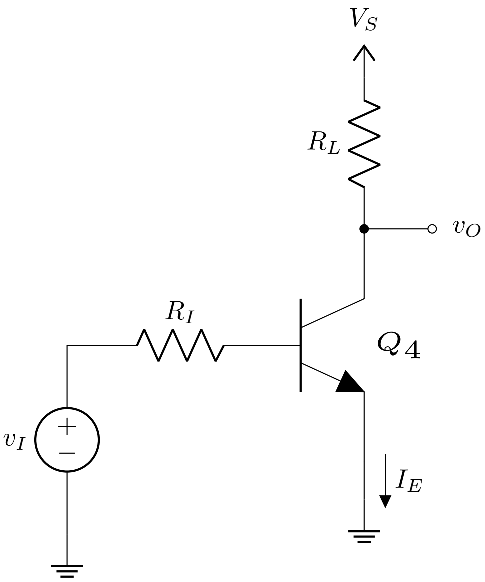
In Saturation: $I_{DS} = \dfrac{k_n}{2}\left(V_{GS} - V_{T_n}\right)^2$

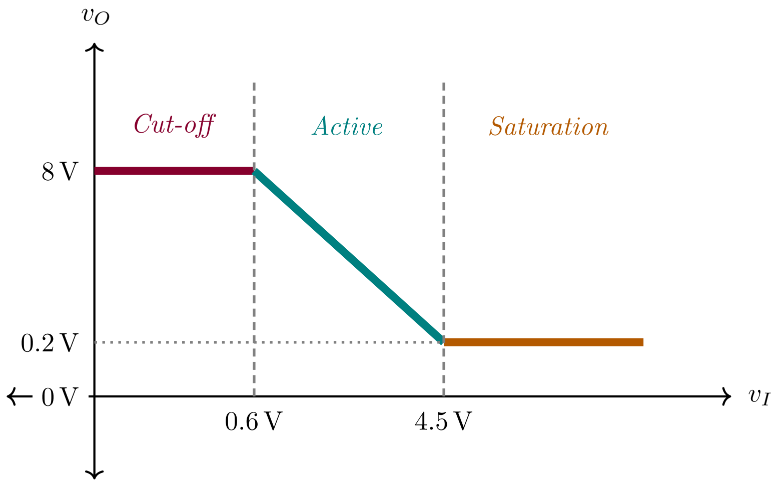
In Triode: $I_{DS} = k_n\Big[\left(V_{GS} - V_{T_n}\right)V_{DS} - \dfrac{1}{2}V_{DS}^2\Big]$

where I_{DS} is the drain-to-source current, and V_{GS} and V_{DS} are the gate-to-source and drain-to-source voltages, respectively.

■ Question 5 of 5 *[CO2] [10 marks]*

The output voltage v_O for the circuit shown below is plotted as a piecewise function of the input voltage v_I. The different regions of transistor operation are indicated by vertical dashed lines, and the specific transistor parameters are provided in the accompanying box:





NPN BJT Parameters:

For the transistor Q_4,

Common emitter current gain: $\beta = 100$

Base-emitter voltage: $V_{BE(sat)} = V_{BE(active)} \approx 0.6\,\text{V}$

Collector-emitter voltage: $V_{CE(sat)} = V_{CE(edge\ of\ sat)} \approx 0.2\,\text{V}$

Analyze the graph and the circuit to answer the questions below:

(a) ***[1 mark]*** Determine the value of V_S.

(b) ***[6 marks]*** When the transistor operates at the edge of saturation, the emitter current is $I_E = 3.94\,\text{mA}$. Determine the values of R_L and R_I. [*Use* $\beta = 100$]

(c) ***[3 marks]*** Using the values found in part (b), what is the value of I_E when $v_I = 5\,\text{V}$?